1. 6 for normal scenario and 5 for congested scenario
2. In normal scenario:
    - Plane-1
    - Plane-2
    - Plane-3
    - Plane-4
    - Plane-5
    - Plane-6

    In congested scenario:
    - Plane-1
    - Plane-2
    - Plane-3
    - Plane-4
    - Plane-5 (Require emergency landing)

    Each thread has sub-threads for:
    - Disembarking and embarking passengers
    - Refilling supplies and cleaning aircraft
    - Refueling
3. Main thread, application worker threads, forked child threads
4. Plane:

```java
public class Plane {
    private static int capacity = 50;
    private int passengers;
    private String name;
    private Airport.Gate gate;
    private boolean emergency = false;

    public Plane(String name) {
        this(name, emergency: false);
        passengers = new Random().nextInt(bound: 50);
    }

    public Plane(String name, boolean emergency) {
        this.name = name;
        this.emergency = emergency;
    }
```

ATC:

```java
public class Airport {
    private static final String RESET = "\u001B[0m";
    private static final String RED = "\u001B[31m";
    private static final String GREEN = "\u001B[32m";
    private static final String YELLOW = "\u001B[33m";
    private int emergencyQueue = 0;

    Runway runway = new Runway();
    RefuelTruck refuelTruck = new RefuelTruck();
    List<Gate> gates = List.of(
        new Gate(name: "Gate A"),
        new Gate(name: "Gate B"),
        new Gate(name: "Gate C")
    );
```

Refuel Truck:

```java
public class Airport {
    private static class RefuelTruck {}
```

Cleaning Service (N/A as a class, instead as a Plane method):

```java
public class Plane {
    public void refillSuppliesClean() {
        String action = String.format(
            format: "%s: Refilling supplies and cleaning...",
            name
        );
        System.out.println(action);

        process(duration: 5000);

        String message = String.format(
            format: "%s: Refilled supplies and cleaned",
            name
        );
        System.out.println(message);
    }
```

Passengers (N/A as a class, instead tracked as number of passengers in a Plane):

```java
public class Plane {
    private int passengers;
```

5.

Object Used in Thread	Instantiated Within Class
Plane	Main
ATC	Main
Refuel Truck	ATC

Plane Member Used in Thread	Appear in Class
Cleaning Service	Main
Passenger	Main

6.

```
public class Main {
    public static Airport airport = new Airport();
    static long maxWait = Long.MIN_VALUE;
    static long minWait = Long.MAX_VALUE;
    static long totalWait; // Average depending on plane count in scenario
    static int totalPassengersEmbarked;
```

7. Gate and Runway Mutual Exclusion
    - Departing plane holds gate lock from the start of undocking and until after take-off, and holds runway during take-off
    - Landing plane holds gate lock from the start of landing until after docking if gate is empty, and holds runway during landing

```java
public void landDockPlane(Plane plane) {
            for (Gate gate: gates) {
                // Hold gate lock until
                // 1. Gate evaluated unavailable
                // 2. Landing and docking on gate
                synchronized (gate) {
                    if (gate.plane == null) {
                        System.out.println(GREEN + String.format(
                            format: "ATC: Assigned %s to %s upon landing",
                            gate.name, plane.getName()
                        ) + RESET);
                        // Hold runway lock until
                        // 1. Landing
                        synchronized (runway) {
                            System.out.println(GREEN + String.format(
                                format: "ATC: Landing permission granted for %s",
                                plane.getName()
                            ) + RESET);
                            plane.land();
                        }
                        plane.dock(gate);
                        completed = true;
                        break;
                    }
                }
            }
```

```java
public void undockDepartPlane(Plane plane) {
    // Hold gate lock until
    // 1. Undocking from gate and departing
    Gate gate = plane.getGate();
    if (gate == null) return;

    synchronized (gate) {
        plane.undock(gate);
        synchronized (runway) {
            System.out.println(GREEN + String.format(
                format: "ATC: Take-off permission granted for %s",
                plane.getName()
            ) + RESET);
            plane.depart();
        }
    }
```

Conditional Synchronization for Emergency and Non-Emergency Planes:

- Non-emergency planes wait until emergency queue is cleared before being able to repetitively search for an empty gate

- Emergency planes notify waiting non-emergency planes to check emergency queue after completing landing

```java
public void landDockPlane(Plane plane) {
    synchronized (this) {
        while (!plane.emergency() && emergencyQueue > 0) {
            try {
                wait();
            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
                return;
            }
        }
    }
```

```java
public void landDockPlane(Plane plane) {
    synchronized (this) {
        if (completed && plane.emergency() && emergencyQueue > 0) {
            emergencyQueue--;
            notifyAll();
        }
    }
}
```

8. Normal Scenario:

   ATC: Processing landing request for Plane-1

   ATC: Assigned Gate A to Plane-1 upon landing

   ATC: Landing permission granted for Plane-1

   Plane-1: Landing...

   Plane-1: Landed

   Plane-1: Docking on Gate A...

   Plane-1: Docked on Gate A

   ATC: Processing landing request for Plane-2

   ATC: Assigned Gate B to Plane-2 upon landing

   ATC: Landing permission granted for Plane-2

   Plane-2: Landing...

   Plane-1: Refueling...

   Plane-1: Refilling supplies and cleaning...

   Plane-1: Disembarking 46 passengers...

Plane-1: Disembarked 46 passengers

Plane-1: Embarking 0 passengers...

Plane-2: Landed

Plane-2: Docking on Gate B...

Plane-1: Embarked 0 passengers

Plane-2: Docked on Gate B

ATC: Processing landing request for Plane-3

ATC: Assigned Gate C to Plane-3 upon landing

Plane-2: Disembarking 2 passengers...

ATC: Landing permission granted for Plane-3

Plane-3: Landing...

Plane-2: Refilling supplies and cleaning...

Plane-2: Disembarked 2 passengers

Plane-2: Embarking 2 passengers...

Plane-1: Refilled supplies and cleaned

Plane-1: Refueled

Plane-2: Refueling...

Plane-1: Undocking from Gate A...

Plane-3: Landed

Plane-3: Docking on Gate C...

Plane-2: Embarked 2 passengers

Plane-1: Undocked from Gate A

ATC: Take-off permission granted for Plane-1

Plane-1: Departing...

Plane-3: Docked on Gate C

ATC: Processing landing request for Plane-4

Plane-3: Disembarking 5 passengers...

Plane-3: Refilling supplies and cleaning...

Plane-3: Disembarked 5 passengers

Plane-3: Embarking 1 passenger...

Plane-2: Refilled supplies and cleaned

Plane-1: Departed

ATC: Assigned Gate A to Plane-4 upon landing

ATC: Landing permission granted for Plane-4

Plane-4: Landing...

Plane-3: Embarked 1 passenger

Plane-2: Refueled

Plane-3: Refueling...

Plane-2: Undocking from Gate B...

Plane-4: Landed

Plane-4: Docking on Gate A...

Plane-2: Undocked from Gate B

ATC: Take-off permission granted for Plane-2

Plane-2: Departing...

Plane-4: Docked on Gate A

ATC: Processing landing request for Plane-5

Plane-4: Disembarking 25 passengers...

Plane-4: Refilling supplies and cleaning...

Plane-3: Refilled supplies and cleaned

Plane-4: Disembarked 25 passengers

Plane-4: Embarking 10 passengers...

Plane-2: Departed

ATC: Assigned Gate B to Plane-5 upon landing

ATC: Landing permission granted for Plane-5

Plane-5: Landing...

Plane-4: Embarked 10 passengers

Plane-3: Refueled

Plane-3: Undocking from Gate C...

Plane-4: Refueling...

Plane-5: Landed

Plane-5: Docking on Gate B...

Plane-3: Undocked from Gate C

ATC: Take-off permission granted for Plane-3

Plane-3: Departing...

Plane-5: Docked on Gate B

ATC: Processing landing request for Plane-6

Plane-5: Refilling supplies and cleaning...

Plane-5: Disembarking 49 passengers...

Plane-4: Refilled supplies and cleaned

Plane-5: Disembarked 49 passengers

Plane-5: Embarking 19 passengers...

Plane-3: Departed

ATC: Assigned Gate C to Plane-6 upon landing

ATC: Landing permission granted for Plane-6

Plane-6: Landing...

Plane-5: Embarked 19 passengers

Plane-4: Refueled

Plane-5: Refueling...

Plane-4: Undocking from Gate A...

Plane-6: Landed

Plane-6: Docking on Gate C...

Plane-4: Undocked from Gate A

Plane-5: Refilled supplies and cleaned

Plane-6: Docked on Gate C

ATC: Take-off permission granted for Plane-4

Plane-4: Departing...

Plane-6: Refilling supplies and cleaning...

Plane-6: Disembarking 19 passengers...

Plane-6: Disembarked 19 passengers

Plane-6: Embarking 12 passengers...

Plane-4: Departed

Plane-6: Embarked 12 passengers

Plane-5: Refueled

Plane-5: Undocking from Gate B...

Plane-6: Refueling...

Plane-5: Undocked from Gate B

ATC: Take-off permission granted for Plane-5

Plane-5: Departing...

Plane-6: Refilled supplies and cleaned

Plane-5: Departed

Plane-6: Refueled

Plane-6: Undocking from Gate C...

Plane-6: Undocked from Gate C

ATC: Take-off permission granted for Plane-6

Plane-6: Departing...

Plane-6: Departed

=== Statistics ===

Gates Empty: true

Minimum Wait: 3004558500

Maximum Wait: 18404768300

Average Wait: 10123937300

Boarded Number of Passengers: 44

Number of Planes Served: 6

Congested Scenario:

ATC: Processing landing request for Plane-1

ATC: Assigned Gate A to Plane-1 upon landing

ATC: Landing permission granted for Plane-1

Plane-1: Landing...

Plane-1: Landed

Plane-1: Docking on Gate A...

Plane-1: Docked on Gate A

ATC: Processing landing request for Plane-2

ATC: Assigned Gate B to Plane-2 upon landing

ATC: Landing permission granted for Plane-2

Plane-2: Landing...

Plane-2: Landed

Plane-2: Docking on Gate B...

Plane-2: Docked on Gate B

Plane-1: Disembarking 4 passengers...

Plane-1: Refueling...

Plane-1: Refilling supplies and cleaning...

Plane-2: Disembarking 15 passengers...

Plane-2: Refilling supplies and cleaning...

Plane-2: Disembarked 15 passengers

Plane-1: Disembarked 4 passengers

Plane-2: Embarking 13 passengers...

Plane-1: Embarking 40 passengers...

ATC: Processing landing request for Plane-5

ATC: Assigned Gate C to Plane-5 upon landing

ATC: Landing permission granted for Plane-5

Plane-5: Landing...

Plane-2: Embarked 13 passengers

Plane-1: Embarked 40 passengers

Plane-5: Landed

Plane-5: Docking on Gate C...

Plane-5: Docked on Gate C

ATC: Processing landing request for Plane-3

ATC: Landing request denied for Plane-3

Plane-5: Disembarking 0 passengers...

ATC: Processing landing request for Plane-4

Plane-5: Refilling supplies and cleaning...

ATC: Landing request denied for Plane-4

Plane-1: Refilled supplies and cleaned

Plane-1: Refueled

Plane-2: Refilled supplies and cleaned

Plane-2: Refueling...

Plane-1: Undocking from Gate A...

Plane-5: Disembarked 0 passengers

Plane-5: Embarking 49 passengers...

Plane-1: Undocked from Gate A

ATC: Take-off permission granted for Plane-1

Plane-1: Departing...

Plane-5: Embarked 49 passengers

Plane-1: Departed

ATC: Processing landing request for Plane-3

ATC: Assigned Gate A to Plane-3 upon landing

ATC: Landing permission granted for Plane-3

Plane-3: Landing...

Plane-5: Refilled supplies and cleaned

Plane-2: Refueled

Plane-5: Refueling...

Plane-2: Undocking from Gate B...

Plane-3: Landed

Plane-3: Docking on Gate A...

Plane-2: Undocked from Gate B

ATC: Take-off permission granted for Plane-2

Plane-2: Departing...

Plane-3: Docked on Gate A

ATC: Processing landing request for Plane-4

Plane-3: Disembarking 0 passengers...

Plane-3: Refilling supplies and cleaning...

Plane-3: Disembarked 0 passengers

Plane-3: Embarking 32 passengers...

Plane-2: Departed

ATC: Assigned Gate B to Plane-4 upon landing

ATC: Landing permission granted for Plane-4

Plane-4: Landing...

Plane-3: Embarked 32 passengers

Plane-5: Refueled

Plane-3: Refueling...

Plane-5: Undocking from Gate C...

Plane-4: Landed

Plane-4: Docking on Gate B...

Plane-5: Undocked from Gate C

ATC: Take-off permission granted for Plane-5

Plane-5: Departing...

Plane-3: Refilled supplies and cleaned

Plane-4: Docked on Gate B

Plane-4: Disembarking 0 passengers...

Plane-4: Refilling supplies and cleaning...

Plane-4: Disembarked 0 passengers

Plane-4: Embarking 7 passengers...

Plane-5: Departed

Plane-4: Embarked 7 passengers

Plane-3: Refueled

Plane-3: Undocking from Gate A...

Plane-4: Refueling...

Plane-3: Undocked from Gate A

ATC: Take-off permission granted for Plane-3

Plane-3: Departing...

Plane-4: Refilled supplies and cleaned

Plane-3: Departed

Plane-4: Refueled

Plane-4: Undocking from Gate B...

Plane-4: Undocked from Gate B

ATC: Take-off permission granted for Plane-4

Plane-4: Departing...

Plane-4: Departed

=== Statistics ===

Gates Empty: true

Minimum Wait: 3002682700

Maximum Wait: 18404768300

Average Wait: 19124338840

Number of Passengers Boarded: 185

Number of Planes Served: 5